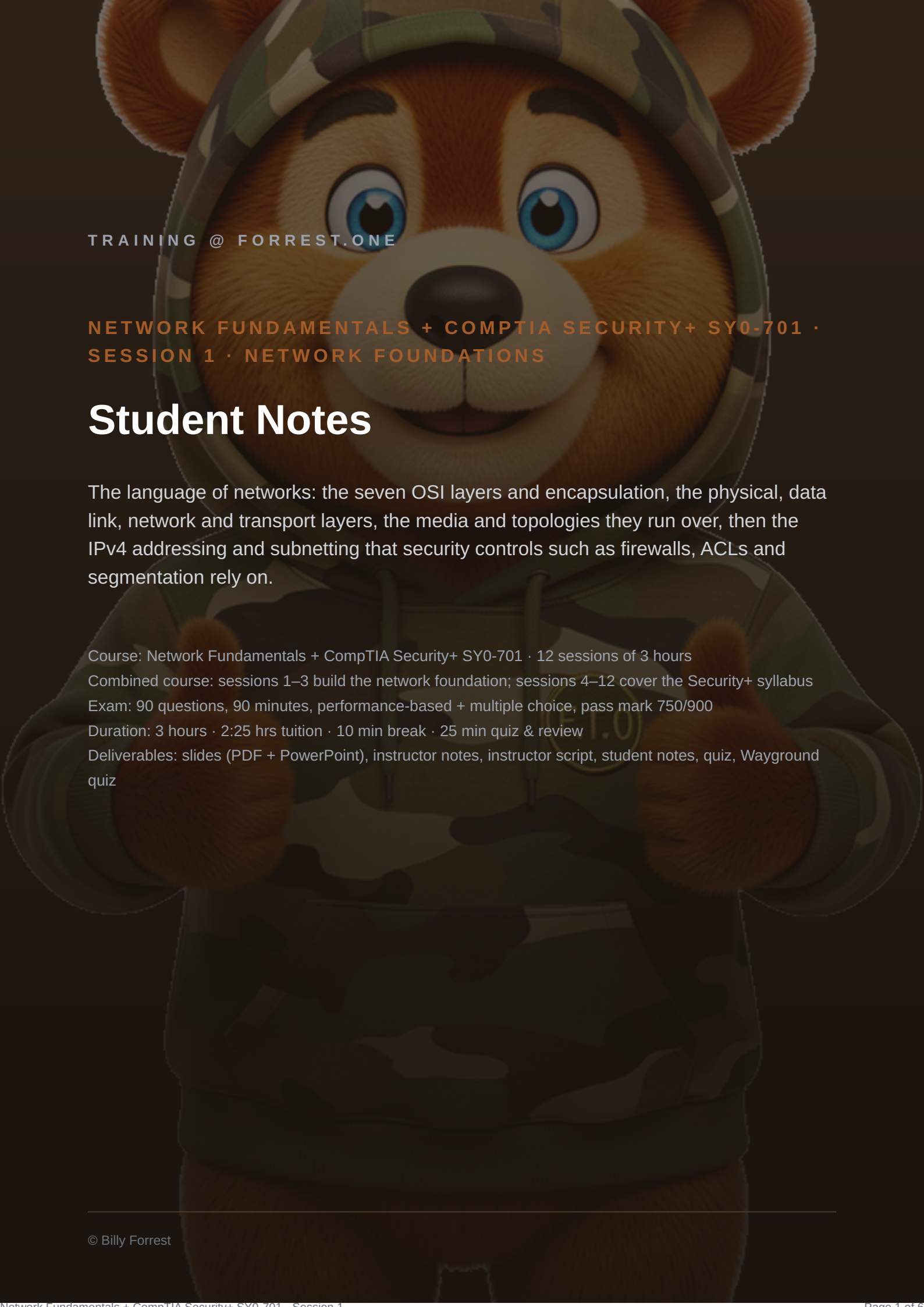




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 1 · NETWORK FOUNDATIONS**

Student Notes

The language of networks: the seven OSI layers and encapsulation, the physical, data link, network and transport layers, the media and topologies they run over, then the IPv4 addressing and subnetting that security controls such as firewalls, ACLs and segmentation rely on.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Combined course: sessions 1–3 build the network foundation; sessions 4–12 cover the Security+ syllabus

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, quiz, Wayground quiz

Student Notes, Network Models & IP Addressing

Your revision guide for Session 1, the first of three network-foundation sessions in the combined course. Read alongside the slides, work the calculations by hand, then use the self-check questions and the session quiz to confirm you are ready for the security content ahead. The session runs in six chapters: the language of networks, the OSI model, physical and data link, network and transport, media and topologies, and IPv4 addressing and subnetting.

Key terms

Term	Definition
OSI model	Seven layers organising network communication, from physical to application
TCP/IP stack	Four-layer model: application, transport, internet, network access
Encapsulation	Each layer adds its header as data travels down the stack
PDU	Protocol data unit: segment or datagram (layer 4), packet (layer 3), frame (layer 2), bits (layer 1)
Port	Transport-layer address, such as 443 for HTTPS and 53 for DNS
MAC address	48-bit hardware address written in hex, used at layer 2, and changeable in software
Logical Link Control	Upper data link sublayer; identifies the layer 3 protocol, error checking
Media Access Control	Lower data link sublayer; shares the medium, delivers by hardware address
Cyclic Redundancy Check	Checksum appended to a frame to detect corruption in transit
Time to Live	Counter in an IP packet that drops by one per router and stops loops
Twisted pair	Cheapest common copper cable; categories rate its capability
Fibre optic	Carries data as light, immune to interference, longest distance
Transceiver	Converts between signal types so a copper port can feed a fibre link
Registered Jack 45	The standard copper connector
Small Form factor Pluggable	Hot-swappable module for changing media type on a switch port
Star topology	Every device connects to a central point: easy to troubleshoot, single point of failure

Term	Definition
Full mesh	Every node links to every other node: maximum resilience, maximum cost
Subnet mask	32 bits, ones then zeros, marking the network/host boundary
CIDR	Classless notation: the slash count of network bits
Private range	10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16: reusable internally, not internet routable
Loopback	127.0.0.1, always the local machine
169.254 range	Self-assigned when no address server answered: a fault symptom
Network address	First address of a range: identifies the subnet, never assigned
Broadcast address	Last address of a range: reaches all hosts, never assigned
VLSM	Variable subnet masks within one network, allocated largest first

Core concepts

Chapter 1, The language of networks

Layering exists so that faults are findable. Without it, a connectivity problem is an unfindable mixture of cable, switch, route and server. With seven layers, each with one job, you test in turn and eliminate as you go. Remember that the model is a map, not a product: nobody sells a layer five, but thinking in layers is how real troubleshooting is done.

Chapter 2, The OSI model

The seven layers are application (7), presentation (6), session (5), transport (4), network (3), data link (2) and physical (1). The TCP/IP stack condenses these to four: application (OSI 5 to 7), transport (4), internet (3), network access (1 to 2). A message travels down the stack: the application produces the data, transport adds ports and produces a segment or datagram, network adds IP addresses and produces a packet, data link adds hardware addresses and produces a frame, and the physical layer transmits bits. The receiver reverses the process. Encapsulation is simply that wrapping, and the vocabulary matters because exam questions use these exact words.

Chapter 3, Physical and data link

Layer 1 carries the raw bitstream with no understanding of meaning. Its three signal types are electrical (copper), light (fibre) and radio (wireless), and it defines the cabling, connectors and signalling standards. Cost, distance and interference decide which medium you use. A completely dead link is almost always a layer 1 fault, so check the cable, the port indicator and the link state before anything else. Layer 2 handles node to node delivery on one local network using 48-bit Media Access Control addresses written in hexadecimal, and its unit of data is the frame. It appends a Cyclic Redundancy Check to detect corruption. It has two sublayers: Logical Link Control, which faces up to the network layer and identifies the layer 3 protocol, and Media Access Control, which faces down to the physical layer, shares the medium and delivers

by hardware address. Hardware addresses are assigned by manufacturers but changeable in software, so they are not a trustworthy identity.

Chapter 4, Network, transport and above

Layer 3 provides logical addressing and routing: a hardware address identifies a device on a local segment, an IP address identifies a device across networks. Routers work at layer 3, Time to Live drops by one per router and discards the packet at zero, which prevents endless circulation, and the layer adapts routes when links fail. Layer 4 delivers between applications using port numbers. Transmission Control Protocol is connection oriented, sequence numbered, guarantees delivery and retransmits losses. User Datagram Protocol is connectionless, unordered and best effort, which makes it faster and lower latency and preferable for voice, video and lookups. Layer 5 manages conversations and checkpoints; layer 6 handles format translation, compression and encryption; layer 7 is where HTTP and DNS live, and it is the protocol layer, not the program.

Chapter 5, Media and topologies

Wireless uses radio waves and buys mobility, but it is a shared medium: lower frequencies travel further and carry less, higher frequencies carry more over shorter distances, and every device on the channel competes for airtime. Wired media: twisted pair is cheapest and most common (Category 5e and Category 6 are the ones you will meet), shielding reduces interference, coaxial suits legacy networks and broadband, and fibre carries light, so it is immune to interference and spans the longest distances. Transceivers convert between signal types, RJ45 is the standard copper connector, SC and LC are the fibre connectors to recognise, and Small Form factor Pluggable modules are hot swappable. Topology is a trade between cost and resilience: star is simple with a single point of failure, full mesh has no single point of failure but is costly and complex, hybrid does both where it matters, and spine and leaf gives predictable latency in the data centre.

Chapter 6, IPv4 addressing and subnetting

An IPv4 address is 32 bits in four octets, written in dotted decimal, each octet 0 to 255, giving about 4.29 billion addresses. Recognise these on sight: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16 are private and not internet routable; 127.0.0.1 is loopback; a 169.254 address is self-assigned when no address server answered and is a symptom, not a configuration. A subnet mask is 32 bits of ones then zeros: the ones mark the network, the zeros the host. Two addresses are always reserved, the network address and the broadcast, so a /24 mask gives 254 usable hosts, not 256. Classless addressing replaced the classes: the slash states the network bits, a larger slash means a smaller network, and /30 gives just two addresses. VLSM lets each subnet be a different size and is allocated largest first, because small-first allocation fragments the space. Worked example: 192.168.1.50 with mask 255.255.255.0 gives network 192.168.1.0, usable hosts 192.168.1.1 to 192.168.1.254, and broadcast 192.168.1.255.

Prefix and mask reference

Prefix	Subnet mask	Usable hosts	Typical use
/24	255.255.255.0	254	Office network
/25	255.255.255.128	126	Half a /24, a large department

Prefix	Subnet mask	Usable hosts	Typical use
/26	255.255.255.192	62	A department or a small server segment
/27	255.255.255.224	30	A small team or a DMZ segment
/28	255.255.255.240	14	A very small segment
/29	255.255.255.248	6	A tiny segment or a management zone
/30	255.255.255.252	2	A point-to-point link between two routers

Practical cautions

- Do not assign the network or broadcast address to a host.
- Do not assume a /24 means 254 usable hosts in every context: it is 254 only when the whole block is one subnet.
- Do not treat a MAC address as an identity: it can be changed in software.
- Do not start troubleshooting a dead link at layer 3: check the cable, the port light and the link state first.
- Do not confuse Logical Link Control, which looks up to layer 3, with Media Access Control, which looks down to layer 1.
- Do not say the application layer is the application: it is the protocol layer, where HTTP and DNS live.
- Do not choose a cable on price alone: distance, bandwidth and interference decide.
- Do not read a 169.254 address as a valid configuration: it means no address server answered.
- Do not allocate VLSM smallest-first: it fragments the address space.
- Do not confuse a larger prefix with a larger network: /30 is two addresses, /24 is 254 usable hosts.

Self-check questions

Answers are in your session slides

If you cannot answer a self-check from memory, revisit the corresponding slide, then retake the quiz.

Exam tips for Session 1

- CompTIA references OSI layer numbers, so know both models and the mapping between them.
- Exam questions use the unit-of-data vocabulary precisely: segment, datagram, packet, frame, bits.
- The subnet formula pattern is the fastest mark on the paper: every extra prefix bit halves the count, and two addresses are always reserved.
- Remember the common connectors: RJ45 on copper, SC and LC on fibre, SFP modules hot swappable.
- Media questions are decided by distance, bandwidth and interference, in that order of frequency.
- A 169.254 address is a fault symptom, and questions use it as a distractor.

- Segmentation is a scored security topic: link subnets to zones, the attack surface and lateral movement.

Further reading

- RFC 791, Internet Protocol (IPv4).
- RFC 4632, CIDR: An Address Assignment and Aggregation Strategy.
- IEEE 802.3, Ethernet, for the data link and physical layer detail.
- IEEE 802.11, Wireless LAN, for the wireless medium.
- CompTIA Security+ SY0-701 Exam Objectives, Domain 3 (Security Architecture).
- CompTIA Security+ SY0-701 Exam Objectives, Domain 4 (Security Operations) for network attack mitigation.